

Drishtant Leuva

+61 432228411 | drishtantl@gmail.com | [LinkedIn](#)

CAREER OBJECTIVE

Data Scientist with 6+ years of experience in machine learning, advanced analytics, and generative AI, specialising in detecting suspicious behavioural patterns, anomalies, and risk signals indicative of fraud, financial crime, and misuse within large-scale datasets. Proven track record in building AI-driven decision systems, predictive models, and data pipelines to support risk-based monitoring, investigation prioritisation, and compliance-driven analytics. Experienced in developing explainable AI solutions, network-based analysis, and real-time analytics frameworks to support risk-based decision-making. Seeking to apply advanced analytics and AI to financial crime surveillance, fraud detection, and risk mitigation.

EDUCATION

University of Technology Sydney, Australia Master of Analytics - Research Passed Awarded International Research Scholarship Award Developed a statistical model for social impact quantification of carbon credit projects and deployed a Gen- AI Auditor for UN SDG verification. Gen-AI Auditor for UN SDG verification	Sep 2022 - Dec 2025
Gujrat Technological University, India. Bachelor of Engineering, Information Technology CGPA 8.25 / 10	Jun 2014 - Jul 2018

PROFESSIONAL EXPERIENCE

University of Technology Sydney Research & Industry AI Role, Sydney, Australia Delivered AI-driven analytics, predictive modelling, and generative AI solutions focused on pattern detection, anomaly identification, and risk-based surveillance systems to identify suspicious behaviour and potential compliance risks across regulatory, sustainability, and enterprise data environments. Built scalable data pipelines and analytical frameworks to support risk-based decision-making and operational intelligence.	Sep 2022 – Jan 2026
Academic Teacher University of Technology Sydney	Feb 2023 – Current

Selected Contributions

- Delivered tutorials and academic support across core IT and business subjects, simplifying complex concepts into practical, real-world applications.
- Designed and facilitated interactive sessions on data-driven decision-making, system design, and enterprise IT strategy.
- Mentored undergraduate students, improving conceptual clarity in databases, analytics, and business process modelling.
- Provided detailed assessment feedback, aligning student work with academic standards and industry expectations.
- Collaborated with subject coordinators to refine weekly teaching materials and ensure curriculum relevance.
- Supported students in applying theoretical concepts (e.g., BI frameworks, IS architecture) to case-based problem solving.

Technical Skills

- Data & BI:** SQL, Power BI, Tableau, Excel
- Programming:** Python, R
- Systems:** Database design, ER modelling, BPMN
- Tools:** MySQL, PostgreSQL, Jupyter Notebook

Selected Contributions:

- Built a GenAI-based tutoring system using LLMs and RAG to analyse unstructured text data and identify patterns, inconsistencies, and contextual signals to support intelligent decision-making, helping users navigate complex software engineering concepts.
- Integrated explainability techniques (SHAP, LIME) to make model outputs more transparent, allowing users and stakeholders to understand how decisions were being made.
- Developed backend pipelines and dashboards to monitor system behaviour and detect unusual patterns and anomalies in user interactions to support early identification of irregular activity, enabling continuous monitoring and improvement.
- Designed structured response workflows to ensure the AI produced consistent, reliable, and interpretable outputs, especially when handling complex or ambiguous inputs.

Tech & Tools: Python, Django, GPT, RAG, PostgreSQL, MongoDB, ChromaDB, SHAP, LIME, REST APIs

Research Associate (Short-Term Contract)

Nov 2024 – Feb 2025

Selected Contributions:

- Conducted analytical research, including literature reviews, statistical analysis, and development of data-driven research methodologies.
- Performed computational data analysis and developed analytical frameworks to support research investigations.
- Delivered stakeholder-focused technical reports and data-driven insights, improving clarity and effectiveness of research outcomes.

Tech & Tools: Python, R, Statistical Analysis, Data Modelling, Research Analytics, Excel

AI & Software Engineering Researcher (Industry Project)

Aug 2023 – Jul 2024

Selected Contributions:

- Built an AI-driven system to automate regulatory compliance checks across multiple Sydney councils, reducing manual assessment effort.
- Developed data pipelines to process and structure complex regulatory documents, turning them into usable datasets for decision-making.
- Applied geospatial analysis and external APIs (Google Maps) to enhance location-based insights and improve accuracy of compliance assessments.
- Designed logic-based workflows to identify inconsistencies, anomalies, and compliance risks through rule-based and data-driven analysis.
- Developed logic-based detection workflows to flag irregular conditions and support compliance monitoring

Tech & Tools: Python (Django), Machine Learning, Google Maps API, Geospatial Analytics, SQL, Data Automation

Business Researcher Placement (BRP)

Jan 2023 – Jul 2023

Selected Contributions:

- Developed a **Change Detection AI model** to automatically identify regulatory updates across councils.
- Automated monitoring of policy and compliance changes using machine learning and structured data processing.
- Transformed unstructured regulatory documents into actionable datasets, improving compliance tracking and decision-making.

Tech & Tools: Python, Machine Learning, NLP, Data Processing

Machine Learning Engineer- Carbon Credit Analytics

Sep 2022 – Sep 2024

Selected Contributions:

- Built predictive and statistical models to analyse large-scale datasets, identifying patterns and key factors influencing pricing and project outcomes.
- Developed scoring frameworks to evaluate project performance and impact, supporting structured decision-making.
- Applied machine learning techniques to improve model accuracy and reliability across datasets.
- Worked with secure and traceable data systems to ensure transparency and integrity of analytical outputs.
- Analysed relationships across entities and data points to uncover hidden patterns, correlations, and potential risk linkages.

- Applied behavioural and pattern-based analysis techniques to identify irregular activities and potential risk signals across complex datasets.
- Applied NLP and LLM techniques to extract insights from unstructured data, enabling improved pattern recognition and anomaly detection

Tech & Tools: Python, Machine Learning, Time-Series Analysis, Statistical Modelling, SQL

eClinicalWorks Pvt Ltd, Ahmedabad, India	Feb 2020 – Jun 2022
Data Scientist	
• Key Responsibilities ○ Built predictive models to detect anomalies, identify suspicious patterns, and highlight potential operational and financial risks and highlight issues like potential fraudulent billing patterns and irregular usage behaviour, irregular system usage, and workflow inefficiencies. ○ Automated data pipelines to process and analyse data faster, reducing manual work. ○ Created Power BI dashboards to present insights clearly to stakeholders and support decision-making. ○ Worked with Azure to deploy and maintain scalable data solutions. ○ Performed behavioural analysis on large datasets to detect deviations from normal patterns and support risk identification • Key Achievements ○ Enabled data-driven product decisions through predictive analytics. ○ Reduced manual reporting through automation. ○ Delivered actionable insights supporting strategic planning. • Tech & Tools: Python, Azure, Machine Learning, Power BI, SQL, Data Visualisation	

Tops Infosolutions, Ahmedabad, India	Dec 2018 – Feb 2020
Web Developer (PHP)	
• Key Responsibilities ○ Developed backend systems and APIs to support scalable web applications. ○ Improved system performance and data handling through optimisation and efficient database integration. • Key Achievements ○ Improved system performance and scalability. ○ Delivered integrated multi-platform solutions. • Tech & Tools: PHP, MySQL, APIs, JavaScript, HTML, CSS	

Vsupport Infotech, Ahmedabad, India	Jul 2018 – Nov 2018
PHP Developer	
• Key Responsibilities ○ Built and maintained web applications, focusing on stability and performance improvements. ○ Assisted in deployment, debugging, and technical issue resolution. • Key Achievements ○ Delivered reliable applications aligned with client requirements. ○ Improved system stability through optimisation. • Tech & Tools: PHP, MySQL, HTML, CSS, JavaScript	

TECHNICAL EXPERTISE

- **Machine Learning & Analytics:** Predictive Modelling, Anomaly Detection, Pattern Recognition, Statistical Analysis, Risk Modelling, Anomaly Detection, Behavioural Analytics, Pattern Recognition
- **AI & GenAI:** LLMs, RAG, NLP, Text Mining, Explainable AI (SHAP, LIME), NLP for Unstructured Data Analysis, LLM-based Pattern Detection
- **Programming:** Python, SQL, R
- **Data Engineering:** ETL Pipelines, Data Transformation, Data Validation
- **Visualisation:** Power BI, Tableau, Dashboarding
- **Cloud & Tools:** Azure, AWS, Git

RESEARCH AND ACADEMIC CONTRIBUTIONS

• International Research Scholarship (IRS) University of Technology Sydney Developed ML and blockchain solutions for carbon credit pricing and social impact analysis, improving model accuracy and enhancing data transparency. • Innovate NSW Grant Department of Industry, Innovation and Science / UTS	Sep 2022 – Sep 2024 Jan 2023 – Jul 2024
--	--

Built AI-driven solutions for stormwater regulatory compliance, including change-detection models and automated geospatial data workflows.

- **Academic Project — Income Tax Investment Model (Final Year Project)** *Dec 2017 – Jul 2018*
Developed a .NET and MySQL-based web application to automate income tax investment planning and decision support.

Publication — Springer Book Chapter
Digital Transformation and Sustainability: Methods and Applications *Dec 2025*
DOI: 10.1007/978-3-031-91278-8
Proposed statistical and GenAI framework for verifying UN SDG claims in carbon credit projects.

Conference Contribution
Exploring Trust in Human-LLM Feedback Systems — Poster Track *Feb 2026*
Studied student trust behaviour in AI-assisted software engineering education.

REFERENCES AVAILABLE ON REQUEST